A Compact Narrative for Task-Visible Operator Mechanisms

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1 The Question

The starting point is a simple ambiguity. A transformer trained for in-context linear regression may make predictions close to linear-kernel ridge regression, but pointwise accuracy on one sampled label vector does not identify the computation being used. Many algorithms can agree with ridge regression on the realized labels while responding differently to nearby counterfactual labels.

The note therefore changes the object of study. Instead of asking only whether the transformer predicts the right targets for one sampled context, it freezes the input geometry and perturbs only the context labels. The first question becomes:

When labels are changed infinitesimally, does the transformer respond like the ridge operator?

This operator-response test is only the entry point. The paper then asks a stronger mechanistic question: if the transformer has a ridge-like local response, is that response mediated by a compact context-side activation subspace, does the subspace have the dimension predicted by the task-visible ridge spectrum, and are the high-leverage directions in that subspace causally used?

The resulting claim is stronger than behavioral imitation but narrower than global equality. In the tested regime, the transformer locally realizes the task-distributed ridge label-response operator; that behavior is explained by a reduced KRR operator induced by activation-derived directions; the number of required directions is predicted before looking at activations; and removing the task-important directions selectively damages the response. The locality matters: the tests concern the sampled geometry, the chosen perturbation scale, and especially the task-label directions rather than all possible label perturbations.

2 Fixed Geometry Makes Ridge Regression an Operator

Fix one in-context episode with context and target inputs

$$X_{\mathrm{ctx}} \in \mathbb{R}^{n_{\mathrm{ctx}} \times d_x}, \qquad X_{\mathrm{tgt}} \in \mathbb{R}^{n_{\mathrm{tgt}} \times d_x}.$$

The fixed geometry is

$$G = (X_{\mathrm{ctx}}, X_{\mathrm{tgt}}).$$

During the operator tests, G is held fixed. The context inputs do not move, the target inputs do not move, and the only varying object is the context-label vector

$$y = (y_1, \ldots, y_{n_{\mathrm{ctx}}}) \in \mathbb{R}^{n_{\mathrm{ctx}}}.$$

Define

$$K = X_{\mathrm{ctx}} X_{\mathrm{ctx}}^{\top}, \qquad K_t = X_{\mathrm{tgt}} X_{\mathrm{ctx}}^{\top}, \qquad A = K + \sigma^2 I.$$

For linear-kernel ridge regression, the dual coefficients and target predictions are

$$\alpha^\star = A^{-1}y, \qquad y^\star = K_t\alpha^\star = K_tA^{-1}y.$$

Thus, once G is fixed, ridge regression is a linear map from context labels to target predictions:

$$T_G := K_tA^{-1}, \qquad T_G : \mathbb{R}^{n_{\text{ctx}}} \to \mathbb{R}^{n_{\text{tgt}}}.$$

The transformer defines a map on the same spaces:

$$F_\theta^G : \mathbb{R}^{n_{\text{ctx}}} \to \mathbb{R}^{n_{\text{tgt}}}, \qquad F_\theta^G(y) = \widehat{y}_\theta.$$

When the geometry is clear, write T and F. The central comparison is not merely $F(y) \approx Ty$ for one y. It is whether the local response of F to label perturbations matches the operator T.

3 Context-Label Directions and Local Operators

A context-label direction is a vector

$$u \in \mathbb{R}^{n_{\text{ctx}}}.$$

It specifies how the labels on the context examples are perturbed:

$$y + \varepsilon u = (y_1 + \varepsilon u_1, \ldots, y_{n_{\text{ctx}}} + \varepsilon u_{n_{\text{ctx}}}).$$

This direction lives in label space over context positions. It is not a direction in feature space, and it is not a direction in target space. If there are 47 context examples, then these directions live in $\mathbb{R}^{47}$.

The ridge map T is linear in y. The transformer map F, even with G fixed, can be nonlinear in the labels. Its local operator at the observed label vector is the first-order label-response map. If F is differentiable, this is the Jacobian

$$J_F(y)u = \frac{d}{dt}F(y + tu)\bigg|_{t=0}.$$

Experimentally, the note estimates this response with the central finite difference

$$D_\varepsilon F(y)[u] := \frac{F(y + \varepsilon u) - F(y - \varepsilon u)}{2\varepsilon}.$$

The behavioral operator test asks whether

$$D_\varepsilon F(y)[u] \approx Tu$$

on held-out label directions.

This must not be overread. Matching the derivative at one point does not imply global equality. For example, $f(x) = x + x^2$ and $g(x) = x$ have the same value and derivative at $x = 0$, but they differ away from zero. The finite-difference test therefore supports a tangent-level claim: for small perturbations in the tested directions,

$$F(y + \delta) \approx F(y) + T\delta.$$

It does not prove that $F(y') = Ty'$ for all nearby or distant label vectors.

4 Why the Additivity Defect Is Needed

The finite-difference response is only useful as an operator diagnostic if it is approximately linear in the probe direction. Ridge regression has the property

$$T(u+v) = Tu + Tv.$$

A nonlinear transformer need not have the analogous local property

$$D_\varepsilon F(y)[u+v] \approx D_\varepsilon F(y)[u] + D_\varepsilon F(y)[v].$$

If this approximate additivity failed badly, then an error such as $E_\varepsilon(F,T)$ would be hard to interpret. A large discrepancy could mean that the transformer is using the wrong linear operator, but it could also mean that there is no stable local linear response at the perturbation scale being tested.

The additivity defect is a guardrail against that ambiguity. For paired directions

$$V = \{(u_j, v_j)\}_{j=1}^m,$$

define the second mixed finite difference

$$C_j = F(y+\varepsilon u_j+\varepsilon v_j) - F(y+\varepsilon u_j-\varepsilon v_j) - F(y-\varepsilon u_j+\varepsilon v_j) + F(y-\varepsilon u_j-\varepsilon v_j),$$

and normalize it as

$$A_\varepsilon(F;V) := \left(\frac{\sum_{j=1}^m \|C_j\|_2^2}{\sum_{j=1}^m \|T(\varepsilon u_j+\varepsilon v_j)\|_2^2 + \eta_{\text{num}}}\right)^{1/2}.$$

For any affine function of the labels, $C_j = 0$ exactly: the four evaluations cancel. Therefore a small additivity defect says that, at the chosen ε and probe law, the transformer's nearby input-output surface is close enough to affine for a linear-operator comparison to be meaningful.

This diagnostic does not identify the operator. A model could have a small additivity defect and still implement the wrong local linear map. The logical role is narrower: first check that the local response is approximately linear, then ask whether that local linear response is the ridge operator or a reduced ridge operator.

5 The Operator Errors

For a held-out probe set $U = \{u_1, \ldots, u_m\}$, define the transformer-to-operator error

$$E_\varepsilon(F,S;U) := \left(\frac{\sum_{j=1}^m \|D_\varepsilon F(y)[u_j] - Su_j\|_2^2}{\sum_{j=1}^m \|Tu_j\|_2^2 + \eta_{\text{num}}}\right)^{1/2}.$$

When $S = T$, this measures whether the transformer's finite-difference response matches exact ridge regression. The denominator normalizes by the ridge-response energy on the same probes, so an error of 0.01 means a root-mean-square discrepancy of about one percent of the typical ridge response size.

For two linear operators, define

$$E(S_1,S_2;U) := \left(\frac{\sum_{j=1}^m \|(S_1-S_2)u_j\|_2^2}{\sum_{j=1}^m \|Tu_j\|_2^2 + \eta_{\text{num}}}\right)^{1/2}.$$

This is used to compare exact ridge T with reduced ridge operators T_Q.

Proposition 1 (Task-probe scale for T_Q)**.** *Let $u = A^{1/2}z$, $z \sim \mathcal{N}(0, I)$, and let $Q^\top AQ = I$. With*

$$C = K_t A^{-1/2}, \qquad W = A^{1/2}Q, \qquad P_Q = WW^\top,$$

the population task-probe error satisfies

$$E_{\text{task}}(T_Q, T)^2 = \frac{\|C(I - P_Q)\|_F^2}{\|C\|_F^2 + \eta_{\text{num}}}.$$

Hence

$$0 \leq E_{\text{task}}(T_Q, T) \leq 1.$$

If W spans the top r right singular directions of C, then

$$E_{\text{task}}(T_Q, T)^2 = \frac{\sum_{i>r} s_i^2}{\sum_i s_i^2 + \eta_{\text{num}}}.$$

Proof. Since $W = A^{1/2}Q$, $W^\top W = Q^\top AQ = I$. For $u = A^{1/2}z$,

$$Tu = K_t A^{-1} A^{1/2} z = Cz, \qquad T_Q u = K_t QQ^\top A^{1/2} z = K_t A^{-1/2} WW^\top z = CP_Q z.$$

Taking expectation over z gives

$$\mathbb{E}\|(T - T_Q)u\|_2^2 = \|C(I - P_Q)\|_F^2, \qquad \mathbb{E}\|Tu\|_2^2 = \|C\|_F^2.$$

Right multiplication by the projection $I - P_Q$ cannot increase Frobenius norm, so the error is at most 1. The singular-value expression is the standard optimal residual for keeping the top right singular subspace of C. □

Proposition 2 (Transformer errors are not bounded by 1)**.** *For a finite probe set U, write*

$$R_U^2 = \sum_j \|Tu_j\|_2^2.$$

If $D_\varepsilon F(y)[u_j] = 0$ for all probes, then

$$E_\varepsilon(F, T; U) = \frac{R_U}{(R_U^2 + \eta_{\text{num}})^{1/2}} \leq 1.$$

If $D_\varepsilon F(y)[u_j] = -Tu_j$, then the same calculation gives a value at most 2. More generally, if $D_\varepsilon F(y)[u_j] = cTu_j$, then

$$E_\varepsilon(F, T; U) = |c - 1| \frac{R_U}{(R_U^2 + \eta_{\text{num}})^{1/2}},$$

so transformer-response errors have no universal upper bound.

Proof. Substitute the three proposed finite-difference responses into the definition of $E_\varepsilon(F, T; U)$. The last expression diverges as $|c| \to \infty$ whenever the ridge response energy R_U^2 is nonzero. □

Proposition 3 (Triangle reading)**.** *For any fixed operator S,*

$$E_\varepsilon(F, T; U) \leq E_\varepsilon(F, S; U) + E(S, T; U).$$

Taking $S = T_Q$ is the decomposition used in the experiments.

Proof. Stack the probe responses as columns. The three numerators are Frobenius norms of $M_F - M_T$, $M_F - M_S$, and $M_S - M_T$, and all have the same denominator $(\sum_j \|Tu_j\|_2^2 + \eta_{\text{num}})^{1/2}$. The result is the Frobenius triangle inequality. $\square$

Remark 1 (Reading the numbers)**.** *For population task-probe reduced operators, $E(T_Q, T) \in [0,1]$. An error e is a relative RMS operator error, while e^2 is the missed fraction of task-probe ridge-response energy. Thus* 0.01 *is about one percent RMS error,* 0.1 *misses about one percent of squared response energy, and* 0.54 *misses about* 29%*. Values near 1 mean the reduced operator leaves almost all task-visible ridge response unexplained.*

For transformer errors, the zero-response baseline is about 1, and values above 1 are worse than having no ridge-like label response on the probes. Very large values mean the local finite-difference response is badly misaligned or unstable relative to ridge. Task-probe and isotropic-probe values should not be conflated: they normalize against the ridge response on different direction laws.

The probe distribution matters. Task-label probes sample

$$u \sim \mathcal{N}(0, A),$$

which matches the conditional covariance of labels under the linear task, up to an overall scale:

$$\operatorname{Cov}(y \mid G) = \tau^2 A.$$

Isotropic probes sample $u \sim \mathcal{N}(0, I)$, testing arbitrary context-label directions equally. The distinction is central: the empirical claim is task-local because the transformer is close to ridge on task-label directions but not on arbitrary isotropic directions.

6 Why the Activation Subspace Is Needed

The behavioral result

$$E_\varepsilon(F, T; U) \text{ small}$$

would show that the transformer locally behaves like ridge regression on the tested directions. But this is still an input-output statement. It does not identify where the computation lives inside the transformer, whether ridge-relevant directions are represented compactly, or whether those directions are causally used.

The activation subspace Q is introduced to move from behavior to mechanism. It is a context-side subspace extracted from context-token activations:

$$Q \in \mathbb{R}^{n_{\text{ctx}} \times r}, \qquad Q^\top A Q = I.$$

Its columns are directions over context positions, made orthonormal in the A-geometry induced by the ridge system. This geometry is not cosmetic: ridge regression solves through $A = K + \sigma^2 I$, so dual/context directions are naturally measured using A.

If the dual ridge solution is forced to lie in $\operatorname{span}(Q)$, the reduced dual vector is

$$\alpha_Q = QQ^\top y,$$

and the corresponding reduced prediction is

$$T_Q y = K_t Q Q^\top y.$$

Thus

$$T_Q := K_t Q Q^\top$$

is the KRR-constrained reduced operator induced by the activation subspace. Saying that Q is sufficient for KRR means

$$E(T_Q, T; U) \text{ is small.}$$

This is not yet a claim that the transformer uses Q. It says that if one performs the reduced ridge computation through Q, the result approximates full ridge on the relevant probes.

The mechanism argument therefore needs three separate checks:

$$E_\varepsilon(F, T; U) \text{ small}, \qquad E(T_Q, T; U) \text{ small}, \qquad E_\varepsilon(F, T_Q; U) \text{ small}.$$

They answer different questions. The first is behavioral: the transformer is locally KRR-like. The second is representational/operator sufficiency: the subspace can support ridge. The third links that subspace to the transformer's actual response.

The separation is forced by the residual decomposition

$$D_\varepsilon F(y)[u] - Tu = \big(D_\varepsilon F(y)[u] - T_Q u\big) + \big(T_Q u - Tu\big).$$

The first term is the transformer-to-reduced-operator residual; the second is the subspace residual. Without both terms, one could make an invalid inference from "the transformer behaves like KRR" and "a subspace exists" to "this subspace explains the behavior."

7 The Task-Visible Rank

Before looking inside the transformer, the note asks a ridge-theoretic question: how many context-label directions are needed to reproduce ridge predictions on labels generated by the task? Under the matched linear task,

$$y = X_{\text{ctx}}\beta + \eta, \qquad \beta \sim \mathcal{N}(0, \tau^2 I), \qquad \eta \sim \mathcal{N}(0, \tau^2\sigma^2 I),$$

the conditional label covariance is

$$\mathrm{Cov}(y \mid G) = \tau^2 A.$$

Thus task-label directions are isotropic after whitening by $A^{-1/2}$. Write a task-label perturbation as

$$u = A^{1/2} z, \qquad z \sim \mathcal{N}(0, I).$$

Then ridge responds as

$$Tu = K_t A^{-1} A^{1/2} z = K_t A^{-1/2} z.$$

This motivates the task-visible operator

$$C_G := K_t A^{-1/2}.$$

Let $s_1(G) \ge s_2(G) \ge \cdots \ge 0$ be the singular values of C_G. These singular values measure how much each task-whitened label direction matters for target prediction. The task-visible rank is

$$r_{\text{task}}(G; \epsilon) := \min\left\{ r : \frac{\sum_{i>r} s_i(G)^2}{\sum_i s_i(G)^2 + \eta_{\text{num}}} \le \epsilon^2 \right\}.$$

This rank is computed before looking at activations. It is the number of task-relevant ridge directions an activation explanation must be able to carry. It is not a raw architectural budget and not a training guarantee.

8 How Activation Subspaces Are Constructed

The subspace Q must be a context-side object because the reduced ridge operator acts on context labels. Context-token activations have exactly the right axis for this. If

$$H_L^{\mathrm{ctx}} \in \mathbb{R}^{n_{\mathrm{ctx}} \times D}$$

is a matrix of final context-token hidden states, then each hidden coordinate defines a length n_{ctx} pattern across context positions. The span of those patterns is the set of context-side directions linearly visible in that activation state.

The extraction step converts such visible patterns into an A-orthonormal trial basis. Given any context-pattern matrix $M \in \mathbb{R}^{n_{\mathrm{ctx}} \times p}$, compute the weighted SVD

$$A^{1/2} M = U \Sigma V^{\top}.$$

Keep the dominant left singular directions, subject to the rank cap and singular-value threshold, and map them back to label coordinates:

$$Q(M) := A^{-1/2} U_{\mathrm{kept}}.$$

Then

$$Q(M)^{\top} A Q(M) = I.$$

The weighting by $A^{1/2}$ is part of the mechanism claim. The ridge dual variables live in the A-geometry, and the reduced operator is $T_Q = K_t Q Q^{\top}$. If the extraction ranked activation patterns in an unrelated Euclidean geometry, the resulting basis would not be aligned with the ridge system being tested.

There are two final-state extractors. The raw extractor takes

$$M_{\mathrm{raw}} = H_L^{\mathrm{ctx}}.$$

It asks a broad availability question: does the final context state contain a context-side subspace from which KRR can be reconstructed? The response-only extractor instead uses hidden-state finite differences:

$$Z_L(u) = \frac{H_L^{\mathrm{ctx}}(y + \varepsilon u) - H_L^{\mathrm{ctx}}(y - \varepsilon u)}{2\varepsilon}, \qquad M_{\mathrm{resp}} = [Z_L(u_1), \ldots, Z_L(u_m)].$$

It asks a stricter question: among the final-state directions that actually move when labels are perturbed, is there a subspace whose reduced ridge operator is close to T?

In both cases the basis is frozen after construction. This matters. If Q were re-extracted for each held-out direction, the test would no longer certify a stable internal object. The empirical claim is instead that one activation-derived subspace, extracted from the unperturbed episode and build probes, induces a fixed operator T_Q that explains held-out responses.

9 The Causal Layerwise Construction

The final-state certificate can show that a useful subspace is present at the end of the computation, but it does not explain how that subspace appears across layers. The layerwise construction is a post-training diagnostic for this question. It looks for context-side directions that satisfy three conditions: they move when labels are perturbed, they are new relative to what earlier layers have already supplied, and they are causally useful for the transformer's response.

At layer ℓ, define the hidden-state finite-difference response

$$Z_\ell(u) := \frac{H_\ell^{\mathrm{ctx}}(y+\varepsilon u) - H_\ell^{\mathrm{ctx}}(y-\varepsilon u)}{2\varepsilon}.$$

Stacking these matrices over build probes gives

$$M_\ell = [Z_\ell(u_1), \ldots, Z_\ell(u_m)].$$

The columns of M_ℓ are context-position patterns that are visible at layer ℓ when context labels move. This is the first filter: a direction that does not appear in label-induced hidden-state changes is not counted as part of the label-computation mechanism.

The second filter prevents double-counting. Suppose earlier layers have already accepted directions. Before processing layer ℓ, the construction forms a pre-existing span

$$R_\ell^{\mathrm{pre}} := \operatorname{span}\{A^t S_j : 0 \le j < \ell,\ 0 \le t \le \ell - j\},$$

where S_j contains the directions first accepted at layer j. The powers of A are an accounting device in the ridge-system geometry. They do not assert that the network literally multiplies by A at each layer; they say that once a direction has appeared, later layers should not receive credit for simple A-geometry refinements of the same direction.

Let Q_ℓ^{pre} be an A-orthonormal basis for this pre-existing span. The novel component of the layer response is

$$N_\ell = \left(I - Q_\ell^{\mathrm{pre}}(Q_\ell^{\mathrm{pre}})^\top A\right) M_\ell.$$

An A-weighted SVD of N_ℓ proposes candidate directions

$$A^{1/2} N_\ell = U_\ell \Sigma_\ell V_\ell^\top, \qquad q_i = A^{-1/2} U_{\ell,i}.$$

These candidates are visible and novel, but visibility is still not use.

The third filter is causal. For a candidate q_i, remove its A-orthogonal projection from the context residual stream at layer ℓ:

$$H_\ell^{\mathrm{ctx}} \longmapsto H_\ell^{\mathrm{ctx}} - q_i q_i^\top A H_\ell^{\mathrm{ctx}},$$

roll the remaining layers forward, and measure the damage to the finite-difference response on a separate causal-probe set. A candidate is accepted only if this damage exceeds a predeclared threshold. This selection rule is deliberately stronger than decodability: a direction must be visible, novel, and functionally consequential at the tested layer.

The accepted directions define the layerwise activation span

$$R_{\mathrm{lay}} := \operatorname{span}\{A^t S_\ell : 0 \le \ell \le L,\ 0 \le t \le L - \ell\}, \qquad d_{\mathrm{lay}} := \dim R_{\mathrm{lay}}.$$

Let Q_{lay} be an A-orthonormal basis of this span. The induced reduced operator is

$$T_{Q_{\mathrm{lay}}} = K_t Q_{\mathrm{lay}} Q_{\mathrm{lay}}^\top.$$

The layerwise test then compares d_{lay} to the task-visible rank and compares $T_{Q_{\mathrm{lay}}}$ to T. Closure defects are reported as a sanity check: they measure how well the accepted span reconstructs held-out hidden responses at each layer. If closure is poor, then the proposed layerwise account is missing much of the hidden response it claims to summarize, even if the endpoint operator happens to look good.

10 Experiment 1: Final-State Operator Certificate

The first experiment asks whether the transformer locally behaves like exact KRR and whether this behavior can be explained through a final context-token activation subspace.

For each episode, the geometry G is fixed, A and $T = K_t A^{-1}$ are computed, and build and held-out probe sets are drawn. Two final-state bases are extracted. The raw basis uses the unperturbed final context hidden-state matrix

$$H_L^{\mathrm{ctx}} \in \mathbb{R}^{n_{\mathrm{ctx}} \times D}.$$

Each hidden coordinate gives a pattern across context positions. An A-weighted SVD of these patterns yields an A-orthonormal basis Q_L^{raw}.

The response-only basis uses finite-difference changes in the final context hidden state:

$$Z_L(u) = \frac{H_L^{\mathrm{ctx}}(y + \varepsilon u) - H_L^{\mathrm{ctx}}(y - \varepsilon u)}{2\varepsilon}.$$

These matrices are stacked over build probes, and the same A-weighted SVD extracts Q_L^{resp}. This basis contains directions that actually move under task-label perturbations.

After extraction, Q is frozen and $T_Q = K_t Q Q^\top$ is evaluated on held-out probes. The main task-probe results are:

$$E_\varepsilon(F, T) = 0.0121.$$

The transformer's local response is therefore close to exact KRR on task-label perturbations. Additivity defects are much smaller than the operator errors: 4.4×10^{-4} on task probes and 3.5×10^{-3} on isotropic probes at $\varepsilon = 10^{-3}$, with the same scale at 10ε. This supports the local-linear interpretation.

The final-state subspace certificate also succeeds. For the raw basis,

$$E(T_Q, T) = 0.00148, \qquad E_\varepsilon(F, T_Q) = 0.01194.$$

For the response-only basis,

$$E(T_Q, T) = 0.00367, \qquad E_\varepsilon(F, T_Q) = 0.01239.$$

Thus the final activation subspaces induce reduced ridge operators close to exact KRR, and the transformer's local response is close to those reduced operators.

The same-span response-fitted low-rank control clarifies what is being certified. It uses the same coordinates $Q^\top u$ but fits a free target-side readout to the transformer's finite-difference responses, rather than using the KRR-constrained map $K_t Q$. On task probes this control fits the transformer slightly better, but it agrees worse with exact KRR. Therefore the reduced operator certificate is not merely saying that some low-rank readout can fit the responses; it is saying that the activation subspace supports the KRR-structured operator.

The limitation appears on isotropic probes. The transformer–KRR error rises to about 0.302. For the raw basis, the reduced operator remains closer to KRR than the transformer response, with $E(T_Q, T) = 0.0525$ and $E_\varepsilon(F, T_Q) \approx 0.292$, but the global isotropic claim clearly fails. Experiment 1 therefore supports a task-local operator certificate, not a global implementation of ridge regression on all label directions.

Table 1: Experiment 1 predictive and behavioral checks, mean over 4 episodes.

Quantity	Transformer	Exact KRR	Transformer–KRR
Pointwise MSE to noiseless teacher targets	0.0161	0.0148	–
Pointwise relative discrepancy	–	–	0.01045
Held-out task operator error $E_\varepsilon(F, T)$	–	–	0.0121

Table 2: Reduced operator vs. same-span response-fitted low-rank control, mean over 4 episodes. $E(F, S)$ denotes transformer-to-S operator error; $E(S, T)$ denotes S-to-ridge operator error.

Probe distribution	Basis	$E(F, T_Q)$	$E(F, T_{\text{actLR}})$	$E(T_Q, T)$	$E(T_{\text{actLR}}, T)$
Task probes	raw	0.01194	0.00726	0.00148	0.01202
Task probes	response	0.01239	0.01169	0.00367	0.01645
Isotropic probes	raw	0.2915	0.2007	0.05254	0.2687
Isotropic probes	response	0.3193	0.2418	0.1278	0.2929

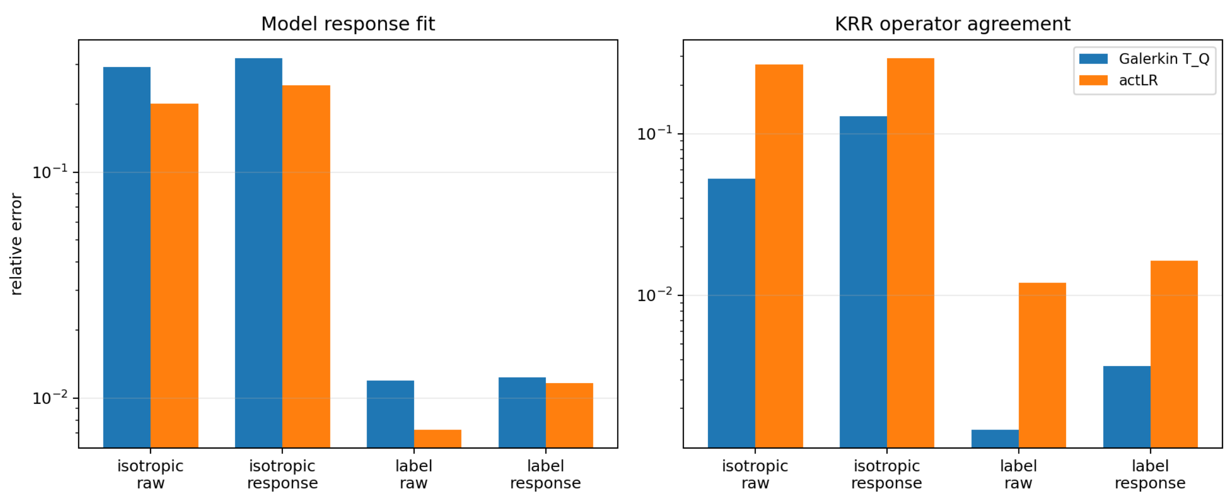

Figure 1: Experiment 1 reduced-operator certificate versus same-span response fitting. The response-fitted operator can fit the transformer slightly better on task probes, but the KRR-constrained reduced operator is much closer to exact KRR.

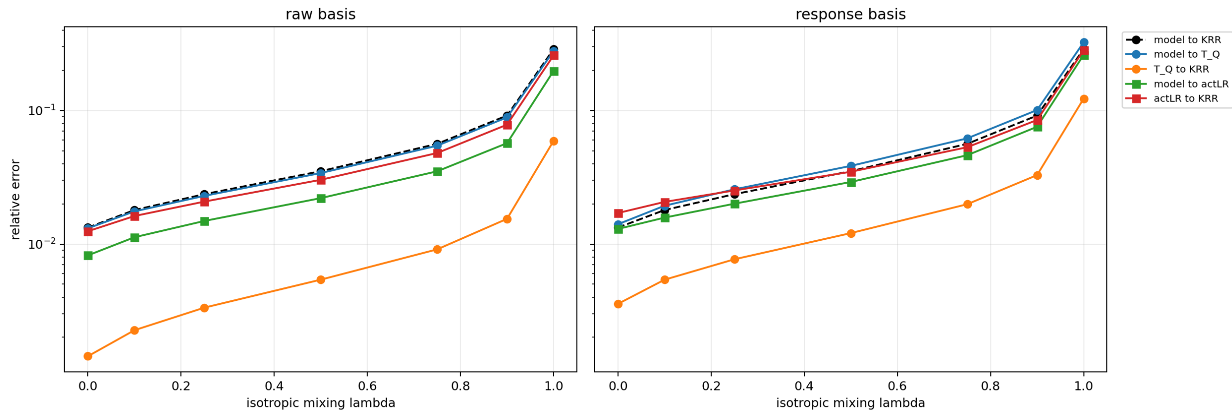

Figure 2: Experiment 1 probe interpolation. The certificate is strong at the task-label endpoint and degrades toward isotropic probes, showing that the claim is task-local rather than global.

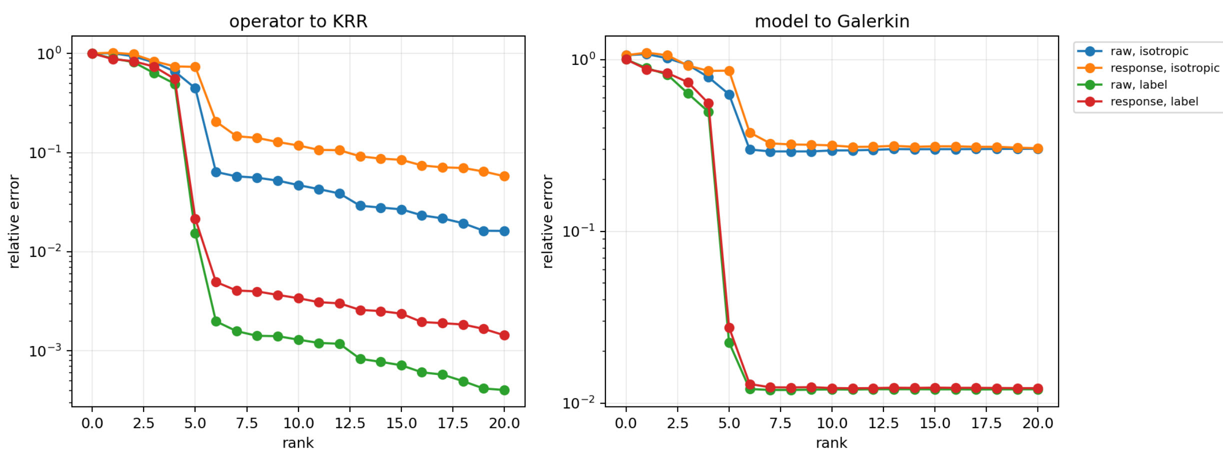

Figure 3: Experiment 1 rank curves for raw and response-only final-state bases. The task-probe operator error falls near the task-visible rank, while isotropic response errors remain much larger.

11 Experiment 2: Layerwise Span and Task-Visible Rank

The second experiment asks how the needed directions appear across layers, and whether their dimension matches the task-visible rank predicted by the KRR spectrum.

At layer ℓ, define the hidden finite-difference response

$$Z_\ell(u) = \frac{H_\ell^{\mathrm{ctx}}(y + \varepsilon u) - H_\ell^{\mathrm{ctx}}(y - \varepsilon u)}{2\varepsilon}.$$

Stacking these responses over build probes gives M_ℓ, a matrix of context-position patterns visible in layer ℓ when labels are perturbed.

The construction processes layers sequentially. Before layer ℓ, directions already accepted from earlier layers define a pre-existing A-orthonormal span Q_ℓ^{pre}. The novel part of the layer response is

$$N_\ell = \left(I - Q_\ell^{\mathrm{pre}}(Q_\ell^{\mathrm{pre}})^\top A\right) M_\ell.$$

An A-weighted SVD of N_ℓ gives candidate directions. Visibility is not enough: a candidate is accepted only if removing it from the context residual stream at that layer changes the transformer's finite-difference response by a predeclared amount.

The accepted directions across layers define a layerwise activation span R_{lay}, with dimension d_{lay}. Let Q_{lay} be an A-orthonormal basis of this span and

$$T_{Q_{\mathrm{lay}}} = K_t Q_{\mathrm{lay}} Q_{\mathrm{lay}}^\top.$$

The hypothesis is a post-training resource law:

$$d_{\mathrm{lay}}/r_{\mathrm{task}} \approx 1 \quad \Longrightarrow \quad E(T_{Q_{\mathrm{lay}}}, T) \approx 0,$$

provided the layerwise closure defects are small.

The target-rank sweep supports this interpretation. For

$$d_x \in \{3, 5, 8, 10\},$$

the selected layerwise dimension reaches the empirical task-visible rank, and the reduced-operator error is numerically zero. For $d_x = 15$, the selected dimension falls slightly short:

$$d_{\mathrm{lay}} \approx 14.5, \qquad r_{\mathrm{task}} \approx 14.9,$$

and the reduced-operator error rises to

$$E(T_{Q_{\mathrm{lay}}}, T) = 0.0997.$$

The failure is interpretable: the selected span often misses one task-visible direction at the largest tested dimension, and the validation closure defect also rises.

The $d_x = 30$ row is an exploratory stress case. Since $n_{\mathrm{tgt}} = 16$, the task-visible rank is capped at $r_T = 16$, even though the input dimension is 30. The same $D = 128, L = 8, H = 4$ architecture and 10k-step training recipe produces a checkpoint that is not behaviorally close to ridge, and the extracted span is not ridge-sufficient despite selecting more than 16 directions:

$$E(T_Q, T) = 0.54023, \qquad E(F, T) = 1.68726.$$

This reinforces the diagnostic separation: accepted causal directions are not enough unless they align with the task-visible KRR directions and the model is in the local KRR-like regime.

Table 3: Experiment 2 complete target-rank sweep with causal layerwise selection, mean over 16 episodes per setting.

Setting	d_{lay}	r_T	d_{lay}/r_T	$E(T_Q, T)$	$E(F, T_Q)$	$E(F, T)$
$d_x = 3$	3.0	3.0	1.000	0.00000	0.00629	0.00629
$d_x = 5$	5.0	5.0	1.000	0.00000	0.00911	0.00911
$d_x = 8$	8.0	8.0	1.000	0.00000	0.01293	0.01293
$d_x = 10$	10.0	10.0	1.000	0.00000	0.01306	0.01306
$d_x = 15$	14.5	14.9	0.971	0.09967	0.10891	0.01743
$d_x = 30$	21.4	16.0	1.336	0.54023	1.56335	1.68726

Table 4: Experiment 2 fixed-$d_x = 5$ depth/head sweep, mean over 16 episodes. All checkpoints use width $D = 128$.

Checkpoint	L	H	d_{lay}	r_T	$E(T_Q, T)$	$E(F, T_Q)$	$E(F, T)$
$H = 1$	8	1	5.0	5.0	0.00000	0.01560	0.01560
$H = 2$	8	2	5.0	5.0	0.00000	0.00904	0.00904
$L = 2$	2	4	20.6	5.0	0.00000	0.18535	0.18535
$L = 4$	4	4	5.1	5.0	0.00000	0.02528	0.02528
$L = 6$	6	4	5.0	5.0	0.00000	0.01255	0.01255
$H = 8$	8	8	5.0	5.0	0.00000	0.00982	0.00982
$L = 8$	8	4	5.0	5.0	0.00000	0.00952	0.00952
$L = 12$	12	4	5.0	5.0	0.00000	0.00944	0.00944

The fixed-$d_x = 5$ depth and head sweeps add an important qualification. These checkpoints keep the task dimension and width fixed while varying depth over $L \in \{2, 4, 6, 8, 12\}$ and head count over $H \in \{1, 2, 4, 8\}$. Causal selection gives $E(T_Q, T) = 0$ for all reported checkpoints, but behavior still varies: the shallow $L = 2$ model has an exact reduced-operator certificate but $E(F, T) = 0.185$, while the $L = 8$, $H = 2$, and $L = 12$ checkpoints have errors near 10^{-2}. Thus the sweep is a control against conflating availability with use:

$$E(T_Q, T) \text{ small}$$

does not imply

$$E_\varepsilon(F, T) \text{ small.}$$

The activations may expose a ridge-sufficient subspace even when the trained transformer does not use it accurately enough to behave like KRR.

12 Experiment 3: Causal Direction Removal

The third experiment asks whether the task-important directions inside Q are actually used by the transformer. This is necessary because decodability and sufficiency do not by themselves prove causal use.

Start with a raw final basis $Q = [q_1, \ldots, q_r]$. For each direction, form Q_{-j} by removing that column. The point of the leverage score is to rank directions by their importance to the KRR-constrained reduced operator, not by activation magnitude. A direction should count as task-important if deleting it changes T_Q's response on task-label probes. Define

$$\Lambda_j^{\text{red}}(U) := \left(\frac{\sum_{u \in U} \|(T_Q - T_{Q_{-j}})u\|_2^2}{\sum_{u \in U} \|Tu\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

A direction has high task leverage if this score is large on task-label probes. This ranking is what connects the causal intervention to the operator certificate: the surgery removes directions that T_Q itself says are important for task-distributed ridge responses.

The intervention removes selected directions from the context-token residual stream:

$$H_s^{\text{ctx}} \longmapsto H_s^{\text{ctx}} - \sum_{j \in J} q_j q_j^\top A H_s^{\text{ctx}},$$

then rolls the remaining layers forward. Matched controls remove low-task-leverage directions from the same basis, high-isotropic-but-low-task directions, high-activation-variance-but-low-task directions, and random subsets.

Before surgery, the checkpoint is in the task-local certificate regime:

$$E_{\text{task}}(F, T) = 0.00926, \qquad E_{\text{task}}(F, T_Q) = 0.00920, \qquad E_{\text{task}}(T_Q, T) = 0.00086.$$

The same episodes retain the off-distribution drift seen in Experiment 1:

$$E_{\text{iso}}(F, T) = 0.266, \qquad E_{\text{iso}}(T_Q, T) = 0.0378.$$

At residual state $s = 1$, removing a single high-task-leverage direction causes prediction drift 4.68, KRR-error inflation 649×, and task-response damage 41.59. Removing a single low-task-leverage direction from the same basis is nearly inert: prediction drift $9.8 \cdot 10^{-4}$, KRR-error inflation 1.03×, and task-response damage 0.0024. The layer sweep localizes the effect: high-task surgery is most damaging at early residual states, before later attention has routed the relevant context information into target tokens, and context-only surgery after the final layer is a no-op. At $s = 1$, aggregate task leverage and task response damage are strongly related across matched non-random controls (log–log Pearson 0.91, Spearman 0.87).

The complement ablation sharpens the mediation claim. At the same residual state $s = 1$, keeping only the A-projection of context residual states onto Q,

$$H_s^{\text{ctx}} \mapsto QQ^\top A H_s^{\text{ctx}},$$

nearly preserves the task-local response over 8 episodes:

$$E_{\text{task}}(F, T) = 0.01866, \qquad \text{response damage} = 0.01659.$$

Removing Q instead,

$$H_s^{\text{ctx}} \mapsto H_s^{\text{ctx}} - QQ^\top A H_s^{\text{ctx}},$$

destroys the response:

$$E_{\text{task}}(F, T) = 145.43, \qquad \text{response damage} = 145.43.$$

So the complement of Q is largely dispensable for the task-local KRR response in this intervention family, while Q itself is necessary.

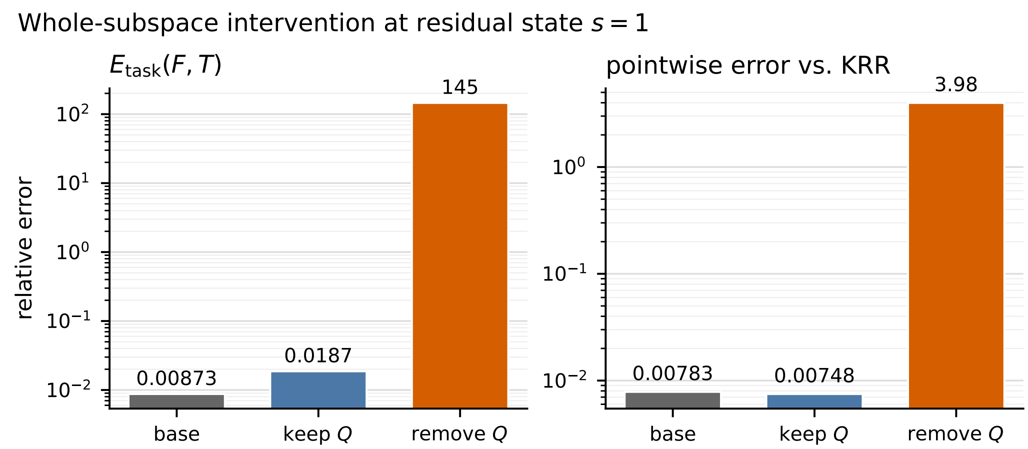

Figure 4: Experiment 3: whole-subspace complement ablation at residual state $s = 1$. Keeping only the Q-projection nearly preserves the task-local response and pointwise KRR prediction; removing Q destroys both.

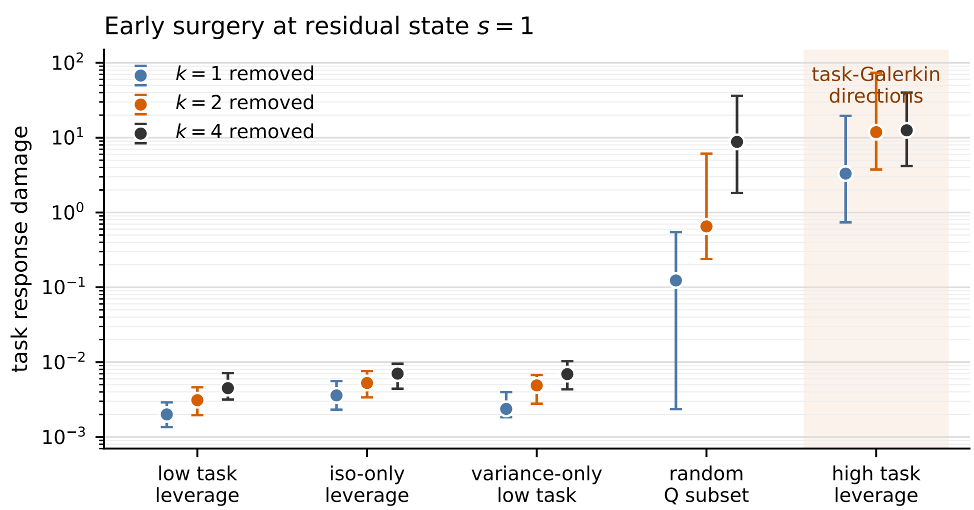

Figure 5: Experiment 3: causal effect of task-important activation directions. Task finite-difference response damage after removing k context-token directions at residual state $s = 1$. High task-leverage directions produce orders-of-magnitude larger damage than low-task-leverage directions from the same activation basis.

This supports the causal-use claim in an operational sense. The directions that the reduced KRR operator identifies as task-important are not merely decodable; removing them selectively damages the transformer's task-local response, while matched controls do not.

13 What Is Justified

The experiments justify a bounded mechanistic claim:

> For the tested checkpoint, geometries, perturbation scale, and task-label probe distribution, the transformer's local label response is well approximated by a reduced KRR operator induced by a compact context-side activation subspace. That subspace reaches the rank predicted by the task-visible KRR spectrum in successful settings, and its high-task-leverage directions are causally important for the transformer's response.

They do not justify stronger claims. They do not show that the transformer globally equals ridge regression. They do not show that derivative matching at one sampled label vector implies equality on a neighborhood. They do not show that the transformer literally executes the explicit algorithm

$$y \mapsto QQ^\top y \mapsto K_t QQ^\top y$$

as its internal program. They also do not show that architecture alone guarantees the behavior. The rank theorem supplies a requirement for context-side subspaces; it is not an SGD theorem.

The clean final formulation is:

> The trained transformer locally realizes the ridge-regression label-response operator on task-distributed directions. In the tested regime, this behavior is explained by a compact context-side activation subspace whose induced reduced ridge operator approximates exact KRR, whose dimension matches the task-visible rank requirement, and whose high-leverage directions are causally important for the transformer's response.

14 Logical Chain

The argument can be read as a sequence of commitments, each narrower than the previous one:

1. Freeze the episode geometry $G = (X_{\text{ctx}}, X_{\text{tgt}})$.
2. Under fixed G, ridge regression becomes $T = K_t A^{-1}$.
3. Treat the transformer as a label map $F : \mathbb{R}^{n_{\text{ctx}}} \to \mathbb{R}^{n_{\text{tgt}}}$.
4. Estimate the local response $D_\varepsilon F(y)[u]$.
5. Test behavior by checking $E_\varepsilon(F, T)$ on task-label probes.
6. Compute the task-visible rank from $C_G = K_t A^{-1/2}$.
7. Extract activation-derived context-side bases Q.
8. Construct the KRR-constrained reduced operator $T_Q = K_t QQ^\top$.
9. Test subspace sufficiency with $E(T_Q, T)$.

10. Test model-subspace alignment with $E_\varepsilon(F, T_Q)$.
11. Compare layerwise span dimension d_{lay} to r_{task}.
12. Remove high-leverage Q-directions and check whether task-local response is damaged.

This is the full mechanism story: behavioral similarity, a KRR-structured activation certificate, rank agreement with the task-visible spectrum, and causal specificity of the high-leverage directions.